A Comparative Analysis of Hydraulic Stability Methods for Rock Protection of Subsea Cables in Nearshore Environments

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Version 1.0
30-6-2026

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1. Introduction

- The rapid growth of the offshore wind sector
- the increasing deployment of subsea cables as critical energy infrastructure
- the intensification of hydrodynamic loading due to climate change.
- Why are existing design methods are not always suitable for nearshore cable protection?
- the gap this thesis addresses.
- research objective and scope of the study.

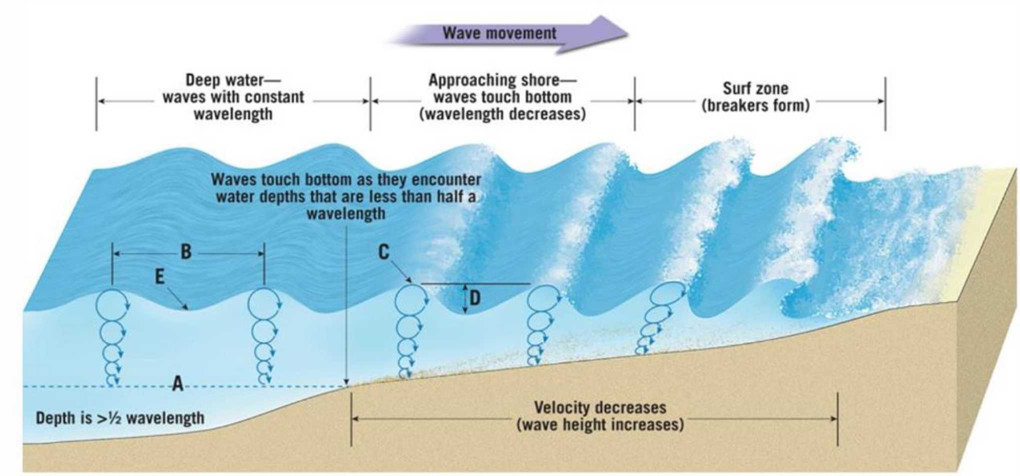
Schiereck & Verhagen (2019) — Ch. 1 (Introduction: why and when, protection against what, failure and design)
Deltares Handbook (2023) — Ch. 1 (Background, considered foundation types) & Ch. 3 (Scour mitigation strategies)

Nowadays we over dimension a lot, especially when a company designs the rock berm when they are responsible. They consider for everything the worst-case scenario, what results in a not sustainable design.

2. The Nearshore Environment

2.1 Definition and area of interest

The coastal region is not a single uniform zone but different areas with different physically difficulties, each characterized by different hydrodynamic conditions and levels of seabed interaction. For the purposes of this study, it is essential to distinguish between three zones: the deepwater/offshore zone, the nearshore zone/approaching zone, and the surf zone (Figure x).



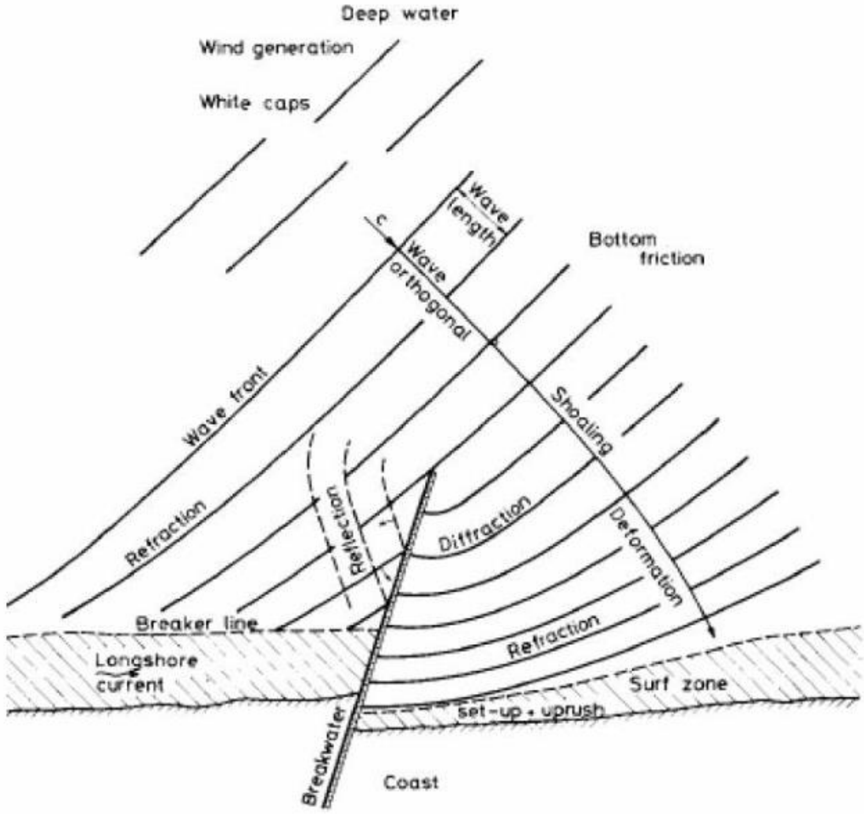
The offshore zone is characterized by water depths sufficiently large that surface waves do not interact with the seabed. In this region, waves propagate largely undisturbed, and the flow conditions above the bed are governed primarily by tidal and large-scale oceanic flows. Wave-induced orbital velocities become negligible at the seabed when the water depth exceeds approximately half the wavelength $(h > L/2)$. Offshore conditions are therefore relatively well understood and predictable, and the design of cable protections in this zone is more straightforward.

The nearshore zone begins where waves start to feel the seabed, where $(h < L/2)$ and extends landward to the point where waves begin to break consistently. This is sometimes referred to as the shoaling zone. In this zone, the interaction between wave orbital motion and the seabed becomes significant: wave celerity decreases, wavelength shortens, and wave height first increases through shoaling before eventually being limited by depth-induced breaking. This zone is the primary focus of this study, as it is precisely here that many nearshore cable routes are laid, and where the complexity of hydrodynamic loading is greatest.

The surf zone extends from the breaker line to the shoreline. In this zone, waves have broken and energy dissipation through turbulence dominates. Longshore currents, undertow, and swash processes are characteristic of this zone. While surf zone dynamics are of interest to coastal engineers, subsea cable routes cannot always avoid the surf zone, for example at landfall sections where the cable must cross it to reach the shore. The surf zone is nonetheless excluded from the scope of this thesis, given the extreme and highly variable loading conditions there, which would require a dedicated treatment beyond what is feasible here. This study therefore focuses on the nearshore zone, with some attention to the transition region near the breaker line where also wave breaking can occur.

2.2 Complexity of Nearshore Environment

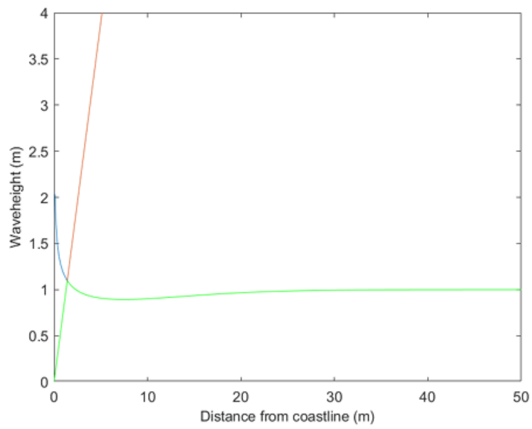
What makes the nearshore zone fundamentally more complex than the offshore environment is the convergence of multiple physical processes acting simultaneously at or near the seabed. The following processes are of direct relevance to the hydraulic stability of rock protection systems.



Shallow water effects. As water depth decreases, the behaviour of waves departs progressively from deep-water linear wave theory. The dispersion relation becomes depth-dependent, wave orbital velocities at the seabed increase significantly, and the wave profile becomes asymmetric, with steeper fronts and flatter troughs. This asymmetry can be explained physically. DNV-RP-F109 (2021, Section 3.4.1) explains that once higher-order wave effects are considered, the near-bed velocity becomes higher under the wave crest and lower (in absolute value) under the trough. This matches the steeper-front, flatter-trough shape described above. The same section also gives a simple check for when linear (Airy) wave theory is still accurate enough, based on wave steepness $H/(gT^2)$ and relative shallowness $d/(gT^2)$. This gives a clear way to check, at any point along the cable route, whether linear wave theory, used in the stability methods of Chapter 5, is still valid, or whether a more advanced wave theory should be used instead.

This asymmetry leads to net sediment transport in the direction of wave propagation, which is an important consideration for the long-term stability of any bed protection. Schiereck and Verhagen (2019) describe how waves travelling from a depth of 20 m towards the shore undergo a complex sequence of shoaling, bottom friction, and eventually breaking, with the relative importance of each process depending strongly on the bed slope.

Wave breaking. Breaking occurs when a wave can no longer sustain its form, either because it becomes too steep in deep water or because the water depth becomes too shallow. The breaking criterion given by Miche provides a theoretical upper limit for wave height as a function of local depth and wavelength (Schiereck & Verhagen, 2019, Eq. 7.8). In practice, for shallow foreshores, a useful approximation is Hb/hb ≈ 0.8–0.88, indicating that the significant wave height is strongly depth-limited in the nearshore zone. DNV-RP-F109 (2021, Section 3.4.1) gives a similar, independently developed breaking criterion: a steepness limit of H/λ ≈ 0.14 in deep water, and a depth limit of H/d ≈ 0.78 in shallow water. This is close to the 0.8–0.88 range given by Schiereck and Verhagen (2019). Because two independent sources agree on a similar depth-limited breaking threshold, this range is used in this thesis as a reasonably reliable criterion for flagging design points that lie within or close to the breaking zone.

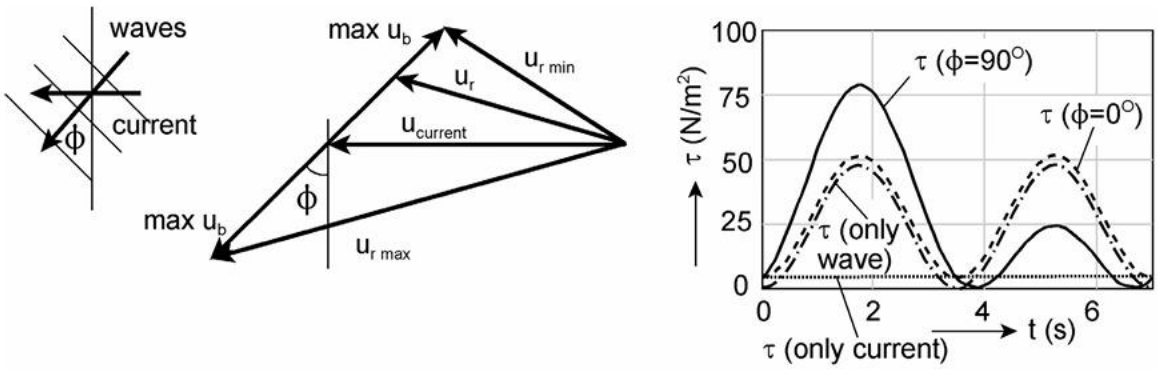


Breaking waves represent a fundamentally different loading category compared to non-breaking waves. Before breaking, the wave field is oscillatory and can be reasonably described by linear wave theory and standard transformation methods. After breaking, the flow becomes turbulent and impulsive, and the impact forces generated are difficult to predict using standard wave theories. For rock protection design, this means that stability formulae derived for non-breaking or mildly breaking conditions may significantly underestimate the actual loading (Herrera & Medina, 2015).

This distinction directly affects which stability method is applicable in this thesis (Chapter 5). Herrera and Medina (2015) specifically address toe berms in very shallow water on steep slopes under breaking wave conditions, making their method one of the few in this comparison validated for the breaking regime. Van Gent and Van der Werf (2014) and Van Rijn (2019), by contrast, are primarily validated for non-breaking or mildly breaking conditions; their applicability must therefore be checked explicitly wherever a design point along the cable route falls within, or close to, the breaking zone. Because the position of the breaker line shifts with the tidal cycle (see below), a single design point may alternate between breaking and non-breaking loading regimes over time.

Tidal influence. Tidal water-level variation chances the breaker line over time, and therefore the boundary between breaking and non-breaking loading, to shift back and forth across the seabed over the tidal cycle. As a result, a protection layer that is stable under non-breaking wave action at high water may be exposed to breaking wave impact at the same location during low water. A better treatment of tidal currents themselves is given in Chapter 3 (Currents), where they are treated as context rather than a governing load.

Wave-current interaction. Waves and currents generally co-exist in the nearshore zone, and their interaction is non-linear: the combined bed-shear stress produced by simultaneous waves and current exceeds the simple sum of each contribution separately, and depends on the angle between wave and current direction (Schiereck & Verhagen, 2019, Section 7.2.2). Consistent with the scope of this thesis, waves are treated as the governing load. A full treatment of combined wave-current loading is kept open as a possible extension (see Chapter 3).



Sediment dynamics. Unlike the offshore zone where the seabed may be relatively stable, nearshore beds are often composed of mobile sediments, fine to medium sands, that are in a state of quasi-continuous transport under the combined action of waves and currents. This has direct consequences for rock protection: sediment can be winnowed (interface instability or piping) out from beneath or between the protection stones. The protection layer can sink into a softer underlying layer, and erosion around the edges of a protection berm can weaken its stability. The Deltares HaSPro Handbook (2023) states that, besides external stability, interface stability and flexibility are two of the three main requirements for offshore rock protections.

- Schiereck & Verhagen (2019)

Ch. 7.3 (Breaking waves, waves on a foreshore)
Ch. 2 (Flow loads, introduction)

- Deltares Handbook (2023)

Ch. 2 (General definitions, general description of the scour process)

- DNV-RP-F109 (2021, Section 3.4.1)

2.3 Bed Morphology and Composition

Subsea cable routes in nearshore environments have a lot of variety of seabed types, each presenting distinct challenges for rock protection design. Two of these conditions are treated as the governing design cases in this thesis: bedrock, where trenching is not possible due to the hardness of the formation, and mobile sandy seabed, encountered mainly at crossings and around obstacles where burial is otherwise not feasible.

Rocky and consolidated beds are commonly found along steep coastlines, especially in fjord areas and around rocky headlands. These seabeds generally carry little risk of winnowing in the strict sense, since there is no loose native sediment beneath the rock surface to be lost. However, the filter and underlayer material of the protection itself can still settle into gaps and fissures in an uneven rock surface over time, a related but distinct interface-stability concern. Their uneven surface can also make it difficult for the protection berm to fully connect with the seabed, which may create unsupported gaps beneath the berm. Herrera and Medina (2015) explain that steep rocky coastlines (with slopes of 1/10 or steeper) behave very differently from gentle sandy slopes, as a result, design formulae developed for sandy coastlines are not directly suitable for these conditions.

Sandy beds are the most common substrate in nearshore zones along low-lying coasts, and become the governing design case at crossings and around obstacles where burial is not feasible. Fine to medium sands (D_{50} in the range of 0.1–0.5 mm) are highly mobile under wave and current action, making them susceptible to scour around and beneath cable protection structures. DNV-RP-F109 (2021, Section 8.4) gives a formal criterion for this mobility: a seabed is considered mobile, or "live", when the bed shear stress from the flow is higher than the threshold shear stress needed to cause significant erosion of the grains present, based on Soulsby (1997). This criterion uses the same type of combined wave-and-current bed shear stress that is introduced later in this thesis for stone stability (Section 4.3). This means the same physical reasoning that decides whether the sand bed itself is mobile also underlies the required stone size of the protection placed on top of it. Sandy beds also present challenges for filter design, as fine sediment can be transported through the voids of the protection layer if the underlayer grading is not properly designed (Schiereck & Verhagen, 2019, Ch. 6), directly relevant to the interface stability requirement of this thesis.

Mixed sand-gravel and gravel beds occur in higher-energy, somewhat steeper nearshore environments, such as exposed Atlantic coasts or tidal inlets, these are generally more stable than pure sand beds but complicate the bed–protection interface due to their coarser, irregular texture. Morphological features such as sand bars, tidal channels, and cross-shore profiles with alternating shallow and deeper areas also occur along cable routes and create local variation in wave breaking, current strength, and bed-shear stress. These conditions are not treated as separate design cases in this thesis but are noted as local variations of the two governing seabed types above, which may require a more detailed spatial assessment along specific sections of the route.

In the nearshore environment, the seabed is often dynamic. Sand waves develop mainly through the interaction between wave orbital motion and the seabed, and can migrate or change shape over time. This makes cable burial planning difficult: a survey carried out two years before installation may no longer represent the actual seabed at the time of construction.

For this reason, seabed conditions and their mobility are assessed on a per-project basis. Based on this assessment, a stable reference level is defined, the Mean Seabed Level (MSBL), which is expected to remain valid over the coming years without becoming exposed. The cable is then buried at a Depth of Lowering (DOL) several meters below this MSBL, giving an additional buffer against future seabed change. The Depth of Cover (DOC) describes the resulting cover above the cable, and the Depth of Trench (DOT), the actual trenching depth, is derived from the required DOL plus the cable diameter and a margin for backfill (NKT HV Cables, "Cable Burial and Protection Plan", Shetland HVDC Link).

Once the required DOL is defined, a trenching tool is selected on a project basis, and its maximum achievable trench depth, Dmax, is checked against this requirement. The trench wall slope must also stay within the tool's capability. Where a seabed feature such as a sand wave locally exceeds the available Dmax, or where its slope is too steep for the trencher, the feature may first be modified by dredging, either fully removed or made gentler and partially infilled, before trenching. If, after these measures, the required depth still cannot be achieved for project-specific reasons, a remedial rock berm is installed to make up the difference and still reach the required protection depth.

A second, distinct situation in which a rock berm is required is where the cable route crosses an existing third-party asset, such as another cable or pipeline. Regulations generally prohibit trenching within such a crossing, so the cable cannot be buried there and instead must be surface-laid. Since the cable cannot be left exposed, a crossing rock berm is installed to provide the required protection depth over this section.

Beyond both of these situations, where the seabed itself is bedrock, trenching is not possible at all. In this case, a rock berm is always required to protect the cable, independent of any depth-lowering shortfall or third-party crossing.

- Schiereck & Verhagen (2019)

Section 7.3 (Breaking waves)
Section 7.3.2 (Waves on a foreshore)
Section 7.2.2 (Shear stress under waves)
Section 2.2 (Turbulence)
Ch. 6 (Filters)

- Herrera & Medina (2015)

Introduction and Section 1 (steep sea bottoms, shallow water)

- Deltares HaSPro Handbook (2023)

Ch. 2 (General definitions)
Ch. 4.2 (Performance requirements: external, interface, flexibility)

2.4 Rock Berm Construction

A rock berm consists of two functional layers. The armour layer, on top, provides external stability, resisting the hydraulic loading from waves and currents (Chapters 4 and 5). Beneath it, a filter or underlayer of finer graded rock provides interface stability, preventing the underlying seabed material from being winnowed through the armour layer, and protects the cable from the impact of dropped armour stones during installation. As a rule of thumb, a filter thickness of four to five rock layers is generally sufficient to prevent winnowing (Deltares HaSPro Handbook, 2023).

For rock material, the EN-13383-1 "Armourstone" standard specifies quality requirements, including limits on the fines content of the material, to reduce turbidity during installation.

In practice, project-specific standards further constrain the geometry. For example, NKT's Cable Burial and Protection Plan for the Shetland HVDC Link specifies a minimum Depth of Cover of 0.6 m and a top width of 1 m for all berms, with side slopes of 1:3 or 1:4 depending on water depth. For remedial rock placement, a minimum berm height of 0.3 m is used, since this is the practical lower limit that a standard installation vessel can achieve with typical offshore aggregate gradings (for example 22-125 mm).

3. Hydrodynamic Forcing

3.1 Currents

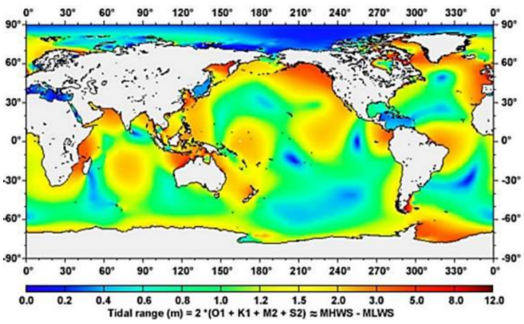
Currents in the nearshore zone are generated by a limited number of physical processes that influence the stability of rock protection along subsea cable routes. In this thesis, the nearshore is defined as the area seaward of the wave-breaking zone, where tidal flow and non-breaking wave motion are the main hydrodynamic processes of interest. These currents are important because they create bed shear stress, transport sediment, and can contribute to scour at the edge of the protection layer. For this reason, understanding the current types present and their typical magnitudes is essential for any stability assessment.

In hydraulic design, flow above a protection layer is often described using the depth-averaged velocity. This is a practical parameter because it is easy to use in calculations and design formulae. However, as Schiereck and Verhagen highlight, the depth-averaged value alone is not enough for protection design. Stone stability also depends on the near-bed velocity, the shape of the vertical velocity profile, and the turbulence intensity close to the seabed. In nearshore conditions, these factors can differ significantly from the depth-averaged flow, especially where tidal flow and wave motion interact.

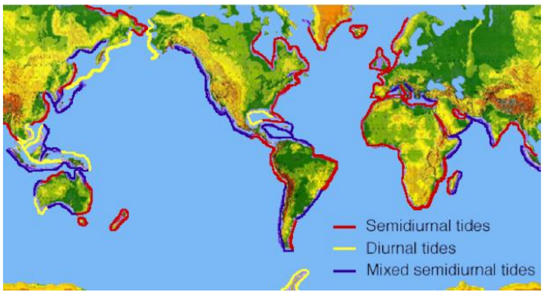
Tidal currents are the most important current mechanism in many nearshore environments and are highly relevant to subsea cable protection design. They are generated by the changing water level caused by the gravitational attraction of the Moon and the Sun acting on the rotating Earth. This astronomical forcing produces a periodic rise and fall of water level, and the horizontal movement of water required to follow that rise and fall creates the tidal current.

The main feature of tidal currents is that they reverse direction during each tidal cycle. During the flood phase, water moves landward. During the ebb phase, it moves seaward. This reversing pattern is important for rock protection because it can cause loading from both directions and may lead to alternating scour on opposite sides of the protection layer. If the flood and ebb phases are not equal, tidal asymmetry may occur, which can produce a net transport of sediment over time and gradually disturb the area around the protection berm.

The magnitude of tidal currents depends on the local tidal range, water depth, and coastal geometry. In some nearshore areas the tidal current is modest, while in other locations it can become strong enough to be one of the governing loads on the protection layer.

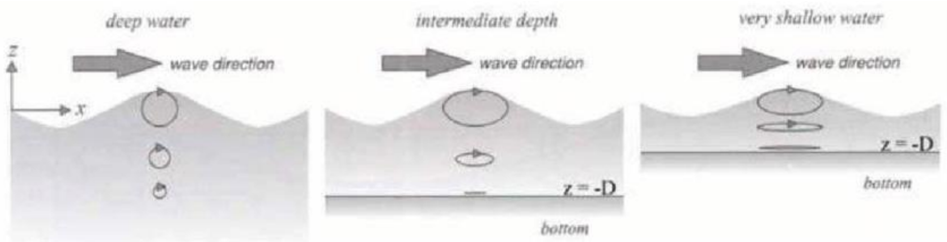


The tidal pattern at a location can also be semidiurnal, diurnal, or mixed, which affects how often the current reverses and how the load is distributed over time: a semidiurnal tide reverses twice a day in a regular pattern, a diurnal tide reverses once a day with longer current phases, and a mixed tide, with unequal flood and ebb, can produce an asymmetric scour pattern around the protection layer.



Wind-driven currents can modify the local current field, particularly when they reinforce the tidal current, though in the nearshore they are generally a secondary contribution that becomes more relevant during storm conditions. Density-driven currents, caused by differences in salinity or temperature, are generally a minor and site-dependent contribution to the total current field.

Wave-induced currents are more directly relevant, since wave motion produces oscillatory flow at the seabed even without breaking, and this motion can interact with the tidal current. These wave-related currents do not act like steady currents but produce repeated oscillatory motion that can destabilise stones, and in nearshore protection design the combined effect of waves and tidal current is often more important than either process alone.



Where waves break at an angle to the shoreline, the resulting alongshore momentum transfer generates a longshore current in the surf zone, whose direction follows the angle of wave incidence at breaking. This current is also the main driver of longshore sediment transport (Battjes, 1974, in Schiereck & Verhagen, 2019). Since the surf zone itself is excluded from the scope of this thesis (Chapter 2), this mechanism is most relevant in the transition region near the breaker line, where it can influence sediment movement around the protection berm.

For protection design, currents should be described in terms of their effect on the rock layer. Near-bed velocity, direction, reversal, and the combined action with waves matter more than the depth-averaged velocity alone. Consistent with the scope of this thesis, the wave parameters used in the stability assessment (Section 3.2) come directly from NKT route data, and waves are generally the largest single contribution to the design load. Current data, provided on a project basis as part of the route data, is nonetheless actively used in this thesis wherever a current is present, both in the combined wave-current stability check (Section 4.3) and, where directional data is available, to establish the angle between wave and current direction, which can meaningfully reduce or increase the combined design load (Section 4.3). Two questions therefore need an answer whenever a current is included: how should the wave and current design conditions be combined, and from what direction do they act relative to each other?

DNV-RP-F109 (2021, Section 2.6) gives a clear answer to this question. A wave or current condition with a stated return period refers here to the characteristic Hs (and associated Tp), or current velocity, that is on average exceeded once during that period, based on long-term statistical analysis of the site's wave and current climate. This is a different statistical concept from the short-term Hs definition introduced in Section 3.2, which describes a single sea state rather than how rare that sea state is over many years. When the joint probability of waves and current is not known, which is normally the case for the wave and current data available for a cable route, the design condition can be approximated by the worse of two combinations: a 100-year wave with up to a 10-year current, or a 10-year wave with up to a 100-year current. The combined load depends on the current-to-wave velocity ratio in a non-linear way. This means a smaller current can sometimes give a more severe load than a larger one, so both combinations should be checked, not just the one with the larger current. For a temporary phase shorter than twelve months but longer than three days, DNV-RP-F109 recommends the same approach but with seasonal 10-year and 1-year return periods instead. For a temporary phase shorter than three days, DNV-RP-F109 allows the extreme load condition to be based on reliable weather forecasts instead of a statistical return period. This return-period combination method is used in this thesis whenever a current is included together with the governing wave load.

3.2 Waves

Waves are generally the largest single hydrodynamic load considered in this thesis, since they generate oscillatory motion throughout the water column and, in particular, near the seabed, though the combined loading from waves together with a current, and the angle between them, is what ultimately governs stability wherever a current is present (Section 4.3). For each location along the cable route, NKT route data directly provide the significant wave height, the peak period, and the wave direction, so these values are used as input to the stability assessment without a separate derivation step. These values are usually already chosen to represent a specific design condition, for example a wave with a 100-year return period for a permanent structure, following standard offshore design practice (DNV-RP-F109, 2021, Section 2.6). Section 3.1 discusses how this design wave condition should be combined with a matching current return period whenever a current is included.

The significant wave height, Hs, is often defined as the average height of the highest one third of the waves in a sea state (H1/3). In practice, however, wave buoys and numerical wave models usually calculate Hs from the wave spectrum instead, as $Hm0 = 4\sqrt{m0}$, where $m0$ is the zeroth-order spectral moment (Schiereck & Verhagen, 2019, Eq. 7.36–7.38). These two definitions give almost the same value for a narrow-banded sea state in deep water. Schiereck and Verhagen (2019) point out that this is no longer true on shallow foreshores, where the highest waves break and the wave height distribution changes shape, exactly the type of nearshore condition considered in this thesis. Hs is still used here as the main measure of wave severity, but it is useful to keep in mind that the value can come from either definition, and that the two are not guaranteed to match closely in shallow, breaking conditions. The stability methods used later in this thesis (Chapter 5) are calibrated directly against Hs rather than the maximum individual wave height in a sea state, so no separate conversion between the two is needed here.

The peak period, Tp, corresponds to the period at the peak of the wave spectrum, representing the frequency band containing the most wave energy, and together with Hs determines the near-bed orbital motion that loads the protection layer. Wave direction is retained as a design parameter because it determines whether wave attack is aligned with or oblique to the cable route, which affects both the local loading and the potential interaction with any current.

DNV-RP-F109 (2021, Section 3.3) gives a practical way to include this effect in the calculation. The wave-induced velocity used in the stability check can be reduced with a direction and spreading factor, before it is applied to a structure that is not perpendicular to the wave direction. This factor is based on a cosine-power spreading function, and its width depends on how spread out the wave energy is around the main direction. Without site-specific data, DNV-RP-F109 recommends using the most conservative value in the range of 2 to 10, with 6 to 8 commonly used for North Sea conditions. This gives a concrete way to reduce the design wave loading for a cable section where wave attack is oblique to the route, instead of only describing the effect qualitatively.

Because the wave climate at the location of interest is already available from the route data, the transformation steps that would otherwise be required, such as shoaling, refraction and numerical propagation with a model such as SWAN, are not needed in this thesis. Wave breaking, which does require separate attention because it changes the loading regime, is treated in Chapter 2.

This chapter has set out the hydrodynamic context for the stability assessment. The wave climate, taken directly from NKT route data, generally represents the largest single contribution to the design load, but currents, and their direction relative to the waves, are actively considered wherever present rather than treated as a simplification to be ignored. Chapter 4 introduces the stability parameters used to translate this loading into a stability criterion, including near-bed flow conditions, and Chapter 5 evaluates the available assessment methods.

- Schiereck & Verhagen (2019), Ch. 7 (Waves), Ch. 2 (Flow loads)
- Van Rijn (2019), Section 2 (Critical bed-shear stress, combined current and wave conditions, Eq. 5)
- DNV-RP-F109 (2021), Section 2.6 (Load combinations), Section 3.3 (Wave directionality and spreading)
- Deltares Handbook (2023), Ch. 2 (Scour prediction, general description of the scour process)

4. Stability Parameters & Criteria

Rock stability is governed by the flow conditions experienced directly at the bed, not by the depth-averaged flow. In uniform, stationary flow, the boundary layer is fully developed and occupies the entire water depth, producing a logarithmic velocity profile with the average velocity located at approximately 0.4 times the water depth above the bed (Schiereck & Verhagen, 2019, Ch. 2). Tidal currents, because of their long period relative to the time needed for the boundary layer to develop, can reasonably be treated as uniform flow for design purposes.

In practice, however, flow in the nearshore is rarely uniform. Accelerations and decelerations, caused for example by structures, bed irregularities, or wave orbital motion, change the thickness of the boundary layer and the shape of the velocity profile: accelerating flow produces a fuller profile with increased near-bed shear stress, while decelerating flow thickens the boundary layer and increases turbulence (Schiereck & Verhagen, 2019, Ch. 2.3, 3.4). This distinction between uniform and non-uniform flow matters because the standard Shields approach (Section 4.1) is strictly valid only for the uniform case.

Turbulence itself is characterised by the turbulent kinetic energy, k, and by the relative turbulence intensity, defined as the ratio of the root-mean-square velocity fluctuation to the local mean velocity (Schiereck & Verhagen, 2019, Ch. 2.2). Elevated turbulence, whether generated by waves, bed roughness, or nearby structures, increases the instantaneous forces on individual stones beyond what the mean velocity alone would suggest, which is why standard, mean-velocity-based stability criteria become less reliable in non-uniform or high-turbulence conditions. This is the basis for the turbulence-based stability parameter introduced in Section 4.2.

4.1 The Shields Parameter (Uniform Flow)

The classical stability criterion for granular material in a current is the Shields (1936) curve, which expresses the critical dimensionless bed-shear stress, or critical Shields mobility number, as a function of a dimensionless particle Reynolds number. For coarse material (D50 > 10 mm), such as the rock sizes considered in this thesis, the critical Shields parameter is approximately constant at θcr,shields = 0.05, independent of the Reynolds number (Van Rijn, 2019).

The Shields data show considerable scatter, partly because the precise definition of "initiation of motion" is not sharply defined and partly because of experimental variability. For design purposes, Van Rijn (2019) represents this scatter with a reduction coefficient, r, applied to the standard Shields value:

$$\theta_{cr} = r \cdot \theta cr, shields \quad with\ \theta cr, shields = 0.05\ and\ r\ in\ the\ range\ of\ 0.4\ to\ 1$$

The reduction coefficient r functions as a damage or safety parameter: a lower value of r corresponds to a more conservative design, while a value closer to 1 corresponds to the classical Shields threshold. This is the mechanism through which an agreed damage level can be translated into a required stone size, and it is used throughout the comparison in Chapter 5.

The Shields approach is reliable for uniform, current-only flow on a horizontal or mildly sloping bed. It becomes less reliable where the flow is non-uniform or where turbulence levels are elevated (Section 4.2), or where waves are present in addition to a current (Section 4.3).

4.2 Turbulence-based Parameters

Hofland (2005, as cited in Van Rijn, 2019) studied the stability of rock in non-uniform flows with relatively high turbulence levels and found that the local mean velocity alone is not sufficient to characterise the loading on individual stones in these conditions. Hofland introduced an additional stability parameter based on the turbulent kinetic energy, k, alongside the local mean velocity, u, to account for the extra destabilising effect of velocity fluctuations that the standard Shields approach does not capture.

This turbulence-based approach becomes necessary wherever the flow departs significantly from the uniform case in Section 4.1: near structures, at bed irregularities, or in the wave boundary layer. Because a free-standing rock berm itself introduces a bed irregularity, and because wave orbital motion in the nearshore is inherently non-uniform, turbulence effects are a relevant consideration for the seabed types and loading conditions in this thesis, alongside the combined wave-current approach of Section 4.3.

4.3 Combined Wave-Current Conditions

Where a current and waves act simultaneously, Van Rijn (2019) shows that the Shields approach remains valid provided the bed-shear stress is computed as the sum of a current-related and a wave-related contribution:

$$\tau b, cw = \tau b, c + \tau b, w \; (Van\ Rijn, 2019, Eq. 5)$$

with:

$\tau b, c = 0.125 \cdot \rho w \cdot fc \cdot (\gamma str \cdot uc)^2$, the bed-shear stress due to current
$\tau b, w = 0.25 \cdot \rho w \cdot fw \cdot (\gamma str \cdot Uw)^2$, the bed-shear stress due to waves

where uc is the depth-averaged current velocity, Uw is the near-bed peak orbital velocity from linear wave theory, and γstr is a velocity and turbulence enhancement factor accounting for the presence of a structure such as a rock berm. The friction factors fc and fw depend on the relative roughness of the bed; Van Rijn (2019) also gives simplified power-function approximations for fc and fw to avoid the iterative solution that the standard formulations require.

This method is directly applicable to the scope of this thesis, since it is, among the five core sources, the only one that explicitly covers combined wave-and-current loading on a bed or berm. The formulation above adds the current- and wave-related shear stress as scalar magnitudes, without considering the angle between the current direction and the wave direction. In practice, many designs conservatively assume waves and current act in the same direction, which maximises the combined shear stress and can lead to unnecessarily large stone sizes, exactly the kind of over-design this thesis aims to avoid (see also Section 5.9, where the same conservative assumption is identified as a weakness in Thusyanthan et al., 2013). Where directional wave and current data are available, for example from a wave rose and current rose derived from the route data, this angle can be used to obtain a less conservative combined shear stress. DNV-RP-F109 (2021, Section 8.4.2.3) gives a directional combination formula for this purpose, based on Soulsby (1997). Applying the same directional logic to the current- and wave-related shear stress terms already defined for Van Rijn's method above gives:

$$\tau_{b,cw} = \sqrt{[(\tau_{b,c} + \tau_{b,w} \cdot cos\theta)^2 + (\tau_{b,w} \cdot sin\theta)^2]}$$

Where θ is the angle between the current direction and the wave direction. When $\theta = 0°$ (same direction), this reduces exactly to the conservative case used above, where the combined stress is highest. As θ increases towards 90°, the combined stress reduces, which can allow a smaller stone size or a different cross-section to be used. This directional refinement is treated in this thesis as an optional enhancement to the basic Van Rijn (2019) approach, to be applied wherever directional wave and current statistics are available for a given point along the cable route

4.4 Decision Framework

The choice between the parameters introduced in Sections 4.1 to 4.3 depends on the hydrodynamic conditions at a given point along the cable route, and this choice directly determines which stability method from Chapter 5 is applicable. This selection logic, including how directional wave and current data is used to refine the combined loading case, is formalised as part of the computational design tool described in Chapter 7

5. Rock Stability Assessment Methods

This chapter compares the six stability assessment methods used in this thesis: Hudson, Van der Meer, Van Rijn, Van Gent and Van der Werf, Herrera and Medina, and the Deltares HaSPro Handbook (RoBeD model). Each method was developed for a specific situation and produces a different type of output, so the six methods are not simply competing options to choose between. Understanding what each method actually calculates, and under which conditions it is valid, is the basis for selecting the right method for a given point along the cable route.

Hudson, Van der Meer, Van Rijn, and Van Gent and Van der Werf all take the site conditions and return a required stone size, together with the damage level that size is calibrated against. Herrera and Medina works the other way around: instead of solving directly for a stone size, it returns the expected damage for a candidate size, so it is used to explore how damage changes as the design is adjusted. The Deltares HaSPro Handbook, through the RoBeD model, goes a step further and produces the full, buildable rock berm geometry together with a deformation class, once a stone size and an accepted damage level have already been chosen. For each method, this thesis considers the physical assumptions it is based on, the required input parameters, the output quantity it produces, the damage level it is calibrated against, and its validity limits. These points are compared directly in Section 5.8.

Correctly matching each method to the situation it was developed for matters in practice, not only academically. Existing rock berm designs are frequently based on a single method applied to every situation, regardless of whether the seabed, slope, or loading conditions actually fall within that method's range of validity. Companies responsible for constructing a rock berm often default to the most extreme assumptions across the board, since the cost of under-design, in the form of claims or failure, is far more visible than the cost of over-design. This is understandable, but it leads to unnecessarily large stone sizes and unnecessarily large rock volumes.

Applying the correct method for each situation, instead of a single conservative method everywhere, allows a rock berm to be sized closer to what is actually needed. This has a direct, practical benefit: less rock material is used, installation takes less vessel time and therefore less fuel, and the resulting design is both more economical and more sustainable, without giving up the required level of protection.

5.1 Hudson Formula

The Hudson formula (Hudson, 1958) is the classical method for the stability of rock armour on a slope under wave attack, and remains widely used as a first estimate. It relates the required stone weight to the wave height, the relative buoyant density, the slope angle, and a stability coefficient KD. KD is essentially a "dustbin factor" that implicitly absorbs the accepted degree of damage, material type, and placement method, and takes different values for different armour units, roughly 3–4 for natural rock (Schiereck & Verhagen, 2019, Ch. 8).

The formula's simplicity comes at a cost: it does not include wave period or a current, and KD is calibrated for approximately 5% damage without giving an explicit damage scale, so results cannot be scaled to other damage levels. Hudson's formula is validated primarily for slope angles in the range 1.5 < $cot\alpha$ < 4, and for waves that do not break at or overtop the toe of the slope. Because it does not represent a bed or berm on a horizontal or mildly sloping seabed, its role in this thesis is limited to a quick, conservative first estimate rather than a design method (Section 5.8).

$$Dn50 \ = \ Hs \ / \ (\Delta \ \cdot \ (KD \cdot cot\alpha)^{1/3})$$

Variable	Meaning	How to determine
Dn50	Required nominal stone diameter (m)	Output
Hs	Significant wave height (m)	Route data
Δ	Relative buoyant density, $(\rho s \ - \ \rho w)/\rho w$	ρs = 2650 or 3000 kg/m³ (variable)
KD	Stability coefficient	Table value, ≈3–4 for natural rock
cotα	Slope gradient	Project/berm geometry

Purpose: a quick, conservative first estimate of armour stone size on a sloped breakwater under wave attack, not intended for a berm on a flat or mildly sloping bed.

Output: the required nominal stone diameter Dn50.

Validity: slopes 1.5 < cotα < 4, non-breaking and non-overtopping waves at the toe, an implicit fixed damage level of roughly 5% with no explicit scale, and no wave period or current included.

5.2 Van der Meer (1988/1998)

Van der Meer (1988) extended the Hudson approach for rubble-mound slope armour under random waves, introducing the wave period, the number of waves, and the notional permeability of the structure, P, as explicit parameters, with separate expressions for plunging and surging breakers. Unlike Hudson's implicit damage level, Van der Meer expresses damage explicitly through the damage number Sd = Ae/Dn50², the eroded cross-sectional area normalised by the square of the nominal stone diameter; a damage level of Sd ≈ 2–3 is generally taken as the start of damage, and failure (where the armour layer is locally removed and the filter becomes exposed) at higher Sd values that depend on the slope (Schiereck & Verhagen, 2019, Ch. 8).

Because the equations are based on many physical model tests, they represent a fitted average of scattered data, with relative standard deviations of about 6.5% (plunging) and 8% (surging) that can be used directly in a probabilistic design. Like Hudson, however, Van der Meer's formulae were developed and validated for breakwater armour slopes, not for a berm on a horizontal or mildly sloping bed, and they do not include a current. Their relevance to this thesis is therefore mainly as a wave-only, slope-based reference point rather than a method for the free-standing berm case (Section 5.8).

$$Plunging\ breakers: Hs/(\Delta \cdot Dn50) \ = \ 6.2 \ \cdot \ P^{0.18} \ \cdot \ (Sd/\sqrt{N})^{0.2} \ \cdot \ \xi m^{-0.5}$$
$$Surging\ breakers: Hs/(\Delta \cdot Dn50) \ = \ 1.0 \ \cdot \ P^{-0.13} \ \cdot \ (Sd/\sqrt{N})^{0.2} \ \cdot \ \sqrt{cot\alpha} \ \cdot \ \xi m^{P}$$

Variable	Meaning	How to determine
Sd	Damage number = $Ae/Dn50^2$	Chosen damage level (start ≈2–3, failure ≈8–17 depending on slope)
N	Number of waves in the storm	From storm duration / Tm
P	Notional permeability	Table value depending on core structure (0 = impermeable, 0.6 = very permeable)
ξm	Iribarren number = $tan\alpha$ / $\sqrt{(Hs/Lm)}$	From slope and wave steepness
Lm	Wavelength at Tm	Linear wave theory

Purpose: sizing armour stone on a rubble-mound breakwater slope under random waves, extending Hudson with wave period, number of waves, and structure permeability.

Output: the required Dn50 at a chosen, explicit damage level Sd.

Validity: slope armour only, not a bed or berm, no current included, separate expressions for plunging and surging breakers, damage from Sd ≈ 2–3 (start) up to higher slope-dependent values (failure).

5.3 Van Rijn (2019)

Van Rijn (2019) extends the classical Shields approach (Chapter 4) specifically to the design of bed and berm protection for large stones, cobbles, boulders, and rocks in the range D50 = 0.03 to 3 m, under a current, waves, or both together. The method uses the critical Shields mobility number, $\theta cr = r \cdot \theta cr,$ shields with θcr,shields = 0.05, where the reduction coefficient r (0.4 to 1.0) sets the accepted level of movement or damage, and combines current- and wave-related bed-shear stress as $\tau b, cw = \tau b, c + \tau b, w$ (Chapter 4, Section 4.3).

Among the five core sources for this thesis, Van Rijn (2019) is the only method that explicitly covers combined wave-and-current loading on a horizontal or mildly sloping bed or berm, rather than a sloped breakwater armour layer, which makes it directly applicable to the free-standing rock berm considered here. It is calibrated against a compilation of field and laboratory data, though this field calibration is comparatively sparse for the largest rock sizes. The method supports both a static design (no movement allowed) and a dynamic design (some permitted movement via the reduction coefficient r), which is useful for the "avoid overdesign" objective of this thesis.

$$\theta cr = r \cdot \theta cr, shields, \theta cr, shields = 0.05, r = 0.4 - 1.0$$
$$\theta = \tau b, cw \,/\, (\Delta \cdot \rho w \cdot g \cdot D50) \leq \theta cr \rightarrow D50 = \tau b, cw \,/\, (\Delta \cdot \rho w \cdot g \cdot \theta cr)$$
$$\tau b, cw = \tau b, c + \tau b, w$$
$$\tau b, c = 0.125 \cdot \rho w \cdot fc \cdot (\gamma str \cdot uc)^2$$
$$\tau b, w = 0.25 \cdot \rho w \cdot fw \cdot (\gamma str \cdot Uw)^2$$
$$Uw = \pi \cdot Hs \,/\, (Tp \cdot sinh(2\pi h/Ls))$$
$$fc = 0.24 \,/\, [log(12h/ks)]^2 \; (simplified: fc, a = 0.11 \cdot (h/ks)^{-0.3}$$
$$fw = exp\{-6 + 5.2 \cdot (Aw/ks)^{-0.19}\} \; (simplified: fw, a = 0.1 \cdot (Aw/ks)^{-0.3}$$
$$Aw = (0.5 \cdot Tp/\pi) \cdot Uw$$

Variable	Meaning	How to determine
uc	Depth-averaged current velocity (m/s)	Current data (Chapter 3), if applicable
h	Water depth (m)	Bathymetry / route data
Ls	Wavelength at significant wave (m)	Linear wave theory, from Tp and h
ks	Nikuradse bed roughness (m)	$ks = \alpha \cdot D50$, α = 1.5–2 for narrowly graded rock
γstr	Velocity/turbulence enhancement factor due to structure	1 without structure, higher with a berm present
ρw	Water density (kg/m³)	≈1025 for seawater
Uw	Orbital velocity	Uw = π·Hs / [Tp·sinh(2πh/Ls)]
Aw	Near-bed peak orbital amplitude (m)	(0.5·Tp/π) · Uw

Purpose: sizing a bed or berm protection (not a sloped breakwater) under current, waves, or both together, the situation central to this thesis.

Output: the required D50, either for a static design (no movement) or a dynamic design allowing some accepted movement via the reduction coefficient r.

Validity: D50 = 0.03–3 m, horizontal or mildly sloping bed, r = 0.4–1.0 sets the damage level, field calibration is comparatively sparse for the largest rock sizes.

5.4 Van Gent & Van der Werf (2014)

Van Gent and Van der Werf (2014) developed a stability formula specifically for the toe structure of a rubble-mound breakwater, based on the characteristic near-bed orbital velocity, û, rather than the wave height directly. The formula explicitly includes the toe berm geometry, the berm width Bt and thickness tt, and uses the damage number Nod (the number of displaced stones within a strip as wide as Dn50) as its damage criterion. It was validated for slopes of 1:1.5 and 1:2, for toe structures within 10% to 30% of the water depth in front of the structure (ht/h), and for a water depth in front of the toe of 1.2 to 4.5 times the wave height (h/Hs); a roughness correction factor is also available for smoother armour slopes.

Because the formula explicitly parameterises berm geometry, it is, of the methods considered here, the closest in form to a free-standing rock berm. However, it was developed and validated for a toe structure that sits below and supports a breakwater armour slope, with wave run-down from that slope acting on the toe; it therefore assumes a slope configuration above the berm that is not present for a cable berm on a bed. Its applicability to the seabed types in this thesis should be treated as indicative rather than direct (Section 5.8).

$$Dn50 \;=\; 0.32 \;\cdot\; Hs \;/\Delta\; (Nod^{\frac{1}{3}}) \;\cdot\; \left(\frac{Bt}{Hs}\right)^{0.1} \;\cdot\; \left(\frac{tt}{Hs}\right)^{\frac{1}{3}} \;\cdot\; \left(\frac{\hat{u}}{\sqrt{(g \cdot Hs)}}\right)^{\frac{1}{3}}$$

Variable	Meaning	How to determine
Nod	Damage number (displaced stones / Dn50 width)	Chosen design value
ht	Water depth above the toe/berm (m)	Bathymetry
Bt	Berm width (m)	Geometry input
tt	Berm thickness (m)	Geometry input
û	Characteristic orbital velocity at the toe (m/s)	Linear wave theory, from Hs, Tm-1,0, h
Tm-1,0	Spectral wave period (s)	From wave spectrum / NKT data

Purpose: sizing the toe structure of a rubble-mound breakwater, using the near-bed orbital velocity instead of wave height directly, and explicitly including berm geometry.

Output: the required Dn50 for a chosen damage number Nod, or Nod for a chosen Dn50.

Validity: slopes 1:1.5 and 1:2 for the armour above the toe, ht/h = 0.10–0.30, h/Hs = 1.2–4.5, waves only, no current, indicative rather than direct for a cable berm since it assumes a supporting armour slope that a cable berm does not have.

5.5 Herrera & Medina (2015)

Herrera and Medina (2015) developed a design formula specifically for toe berms in very shallow water on steep sea bottoms (slope m = 1/10), under breaking wave conditions, precisely the situation for which Herrera and Medina show that existing formulae, including Van Gent and Van der Werf (2014), do not perform reliably. Their formula (Eq. 12) predicts the damage number Nod for a standard toe berm geometry (Bt = 3·Dn50, tt = 2·Dn50), accompanied by 90% confidence intervals. Three damage levels are defined, following the CIRIA Rock Manual (2007): Nod = 0.5 (start of damage, no significant movement), Nod = 2.0 (moderate damage, berm still supporting the armour), and Nod = 4.0 (failure).

The method is valid within the range:

$$-0.15 < hs/Hs0 < 1.5$$
$$-0.5 < hs/Dn50 < 5.01$$
$$0.008 < s0p\ (wave\ steepness) < 0.08$$
$$0.81 < Ns < 3.36$$

for the standard berm geometry and slope m = 1/10 only. Because it is one of the few methods in this comparison explicitly validated for breaking waves on a steep bottom, it is directly relevant to the bedrock case in this thesis, particularly where the cable route lies within or close to the wave-breaking zone discussed in Chapter 2.

$$Nod \approx \left(\frac{(Hs0 \cdot L0p)^{0.5}}{\Delta D_{n50}} - 5.5\right) \cdot \left[\left(-0.2 \cdot \frac{hs}{D_{n50}} + 1.4\right) \cdot exp\left(0.25 \cdot \frac{hs}{D_{n50}} - 0.65\right)\right]^{\frac{1}{0.15}}$$

Variable	Meaning	How to determine
Hs0	Deep-water significant wave height (m)	Offshore wave data
L0p	Deep-water wavelength at Tp = g·Tp²/2π (m)	From Tp
hs	Water depth at the toe (m)	Bathymetry
Δ	Relative buoyant density, $(\rho s - \rho w)/\rho w$	ρs = 2650 or 3000 kg/m³ (variable)
Dn50	Nominal stone diameter, used as input, not solved for directly	Candidate value to test

Purpose: assessing toe berms in very shallow water on steep, breaking-wave coastlines (slope m = 1/10), the one case where the methods above are shown not to work reliably.

Output: unlike the four methods above, this formula does not solve for Dn50 directly. It takes a chosen candidate Dn50 as input and returns the resulting damage number Nod, so it is used the other way around, to explore the trade-off rather than to size the stone directly.

Validity: narrow ranges (-0.15 < hs/Hs0 < 1.5, -0.5 < hs/Dn50 < 5.01, 0.008 < s0p < 0.08, 0.81 < Ns < 3.36), slope m = 1/10 and the standard berm geometry only, breaking waves only.

5.6 Deltares HaSPro Handbook (2023)

The Deltares HaSPro Handbook (2023) represents the current state of the art for offshore scour and cable protection design, and is, of the five core sources, the only one developed explicitly for cable-cover berms. It distinguishes three separate performance requirements that a rock protection must satisfy: external stability (resistance to hydraulic loading), interface stability (preventing the underlying sand from being winnowed through the protection layer), and flexibility (the ability of the protection to accommodate bed-level changes without losing effectiveness).

For external stability, the handbook introduces a rock berm deformation model (RoBeD) that predicts the expected deformation of a berm directly, rather than only a required stone size. Reshaping becomes relevant for higher mobility numbers, above a Shields number of roughly 0.06, and the model defines four deformation classes (0 to 3), from no deformation (class 0, mobility below a threshold value everywhere) to complete wash-away of the profile (class 3, mobility above the threshold even at the flat seabed). The Deltares HaSPro Handbook (2023) recommends a mobility threshold, MOBthresh, of 0.95 for the RoBeD model itself, based on its own validation against physical model tests of rock berms. This is distinct from the handbook's separate fixed mobility-limit tables, which are specific to monopile scour protection (parameterised by pile diameter and KC-number). A rock berm is a fundamentally different structure, so these tables are not transferable here. For a berm, the RoBeD deformation model with its recommended MOBthresh = 0.95 is the appropriate approach instead.

Because it is calibrated for offshore conditions and assumes a wave-current direction perpendicular to the structure, and because it explicitly addresses berm geometry, interface stability, and flexibility together, it is the most directly applicable design reference for the framework developed in this thesis (Chapter 7), alongside Van Rijn (2019) for the underlying stone-size calculation.

Beyond the qualitative deformation classes, the handbook provides an explicit calculation procedure for the required Top of Product Cover, TOPC, the installed rock cover needed so that, after reshaping under the design hydraulic exposure, the remaining cover does not fall below the Minimum Top of Cover, MTOC. The ratio between the bottom width and height of the reshaped (parabolic) berm profile is given empirically, fitted from physical model tests, as a function of the Shields mobility parameter θ:

$$Bberm/hberm = f(\theta) = 6000\theta^2 + 50\theta + 2.4$$

The installed and minimum berm heights are related to the cover requirements via a cable-position offset, here denoted $\Delta cable$ to avoid confusion with the relative density Δ used elsewhere in this thesis:

$$hberm = TOPC + \Delta cable,$$

$$hmin = MTOC + \Delta cable, with\ \Delta cable = Dcable\ for\ a\ surface - laid\ cable,$$

$$or\ \Delta cable = ZDOL - ZSBL\ for\ a\ free - spanning/buried\ cable.$$

The cross-sectional area of the installed (trapezoidal) berm and of the minimum reshaped (parabolic) berm are:

$$Aberm \; = \; hberm \cdot WTOP \; + \; hberm^2 \cdot \alpha$$
$$Amin \; = \; (2/3) \cdot Bberm \cdot hmin$$

Setting the installed area equal to the minimum reshaped area gives a quadratic equation for hberm, which, after taking the maximum root and substituting the relations above, yields the required installed berm height and, finally, the required TOPC:

$$hberm \; = \; 0.5 \cdot \left[\sqrt{((WTOP/\alpha)^2 \; + \; (4/\alpha)(2/3) \cdot f(\theta) \cdot hmin^2)} - \frac{WTOP}{\alpha} \right]$$
$$TOPC \; = \; 0.5 \cdot \left[\sqrt{(WTOP/\alpha)^2 \; + \; (4/\alpha)(2/3) \cdot f(\theta) \cdot (MTOC + \Delta cable)^2} \; - \; WTOP/\alpha] \right] - \Delta cable$$
$$f(\theta) \; = \; 6000\theta^2 \; + \; 50\theta \; + \; 2.4$$

Variable	Meaning	How to determine
θ (θc)	Shields mobility parameter for the design hydraulic exposure	θc = τc / [(ρs - ρw)·g·d50]
TOPC	Required Top of Product Cover, i.e. installed rock cover (m)	Output
MTOC	Minimum Top of Cover, minimum acceptable cover after reshaping (m)	Project requirement / input
WTOP	Top width of the installed berm (m)	Geometry input
α	Side slope parameter, slope expressed as 1:α	Geometry input
Δcable	Cable-position offset (m)	Dcable for surface-laid; ZDOL - ZSBL for buried/free-spanning
Dcable	Outer diameter of the cable (m)	Cable specification

Purpose: finalising the buildable berm geometry, once a stone size and an accepted damage level have already been chosen from the methods above.

Output: the installed berm geometry (TOPC, berm height and width) and a deformation class (0–3), together with a check on interface stability and flexibility.

Validity: relevant for the reshaping regime above θ ≈ 0.06, assumes the wave-current direction is perpendicular to the structure, calibrated for offshore conditions

5.7 Rock Density

Rock density, $\Delta = (\rho s - \rho w)/\rho w$, is treated in this thesis as an explicit design variable rather than a fixed material property, comparing a normal-density preset (ρs = 2650 kg/m³) against a high-density preset (ρs = 3000 kg/m³). The Deltares HaSPro Handbook (2023) shows that higher-density rock is less mobile than normal-density rock of the same grading, because the same mass is achieved with smaller stones that present a smaller area to the hydrodynamic loading; using its RoBeD deformation model, Deltares finds that a normal-density (ρs ≈ 2650 kg/m³) OSG45/180mm grading and a smaller OSG22/125mm grading achieve a similar degree of deformation once the density of the smaller grading exceeds roughly 3000 kg/m³.

This has a direct, practical consequence for the optimisation objective of this thesis: a higher rock density can allow a smaller stone size, and potentially a smaller layer thickness and total rock volume, for the same accepted level of damage. Since the design procedure itself (Sections 5.1–5.6) does not change with density, only Δ, comparing designs on normal- and high-density presets is treated in this thesis as a direct trade-off between material availability (normal-density rock is generally easier to source) and total tonnage, rather than as two separate design methods.

5.8 Comparative Summary

Method	Situation it was made for	Required inputs	Output	Damage criterion	Validity limits	Fit for nearshore cable berm
Hudson (1953/59)	Slope armour, permeable-core breakwater, regular waves	Hs, KD, Δ, cotα	W50 / Dn50	Implicit fixed (~0–5%), no scale	Slope armour, non-overtopped, permeable core (KD=4 non-breaking / KD=2 breaking; impermeable core only with KD=1); deep to shallow water but NOT very shallow (Hs-toe < 70% Hso); no strict cotα range published (KD tables ~1:1.5–1:3); no Tp; no N; no P; no current	Low – first estimate only
Van der Meer (1988/98)	Slope armour, random waves, deep water	Hs, Tm (sm), Δ, Dn50, cotα, P, N, Sd	Dn50 / Ns at a chosen Sd	Sd = Ae/Dn50² (start 2–3, failure ≈8–17)	Deep water (h > 3·Hs-toe); slope armour; cotα = 1.5–6; P = 0.1–0.6; N < 7500; som = 0.01–0.06; ξm = 0.7–7; Δ = 1–2.1 (higher Δ→cotα≥2); separate plunging/surging; no current. Valid in the outer/deep part of the domain, drops out as waves approach breaking	Low–med – slope, not berm/bed
Van Rijn (2019)	Horizontal/mild-slope bed protection, current ± waves	uc, Hs, Tp, h, Δ, slope, α(ks), r, γstr	required D50; + Nmr/Sd (Cheng 2002) for dynamic	r = 0.4–1.0 → θcr = r·0.05 (static); Cheng bed-load (dynamic)	D50 = 0.03–3 m; horizontal/mild slope; roughness h/(αD50) & Aw/(αD50) = 10–300; D50 is sieve size (D50 ≈ 1.2·Dn50); sparse field calibration for largest sizes	High – only one with combined wave+current ; static & dynamic
Van Gent & Van der Werf (2014)	Toe of breakwater armour slope, waves	Hs, Tm-1,0, ht, Bt, tt, Dn50, Δ, cotα, roughness	Nod or Dn50	Nod (displaced stones in a Dn50-wide strip)	ht/h = 0.10–0.30; h/Hs = 1.2–4.5; slopes 1:1.5 & 1:2; wave steepness sp = 0.015–0.04; waves only; non-overtopped; tested on fixed bed (no toe scour)	Med – berm geometry fits, but assumes armour slope above + down-rush
Herrera & Medina (2015)	Toe berm, very shallow + steep (m=1/10), breaking	hs, Hs0, L0p (Tp), Δ, Dn50 (std Bt=3Dn50, tt=2Dn50)	Nod (Eq. 12) + 90% CI	Nod = 0.5 start, 2.0 moderate, 4.0 failure (CIRIA 2007)	-0.15 < hs/Hs0 < 1.5 (negative = emerged berm); -0.5 < hs/Dn50 < 5.01; steepness 0.008–0.08 (Herrera Table, p.10); Ns = 0.81–3.36; m = 1/10; standard berm (Bt=3Dn50, tt=2Dn50); breaking waves only	Med–high for bedrock/steep breaking; fixed geometry, waves only
Deltares HaSPro / RoBeD (2023)	Loose-rock scour protection & cable-cover berms, offshore, current+waves	Shields θ (→ uc, Hs, Tp, h, Δ, D50), WTOP, MTOC, Dcable, side slope	berm geometry (Bberm/hberm, TOPC) + deformation class; external/interface/flexibility	MOBthresh = 0.95 recommended (case-specific allowed); classes 0–3; reshaping/wash-away at θ > ~0.06	reshaping regime; MOB = θ/θcr (App A); θcr Soulsby Eq A.6 (incl. 1.2 factor); ks = 2.5·D50; perpendicular wave+current assumed; offshore-calibrated	High – only one explicitly for cable berms + 3 requirements

Situation	Primary method	Supporting / check	Why
Bedrock, steep nearshore, breaking waves	Herrera & Medina (2015)	Van Gent (range check); Hudson (first estimate)	Only method for steep bottom + breaking; demonstrates that gentle-slope formulas fail here.
Mobile sand, combined waves + current (crossing/obstacle)	Van Rijn (2019)	Deltares RoBeD (deformation + interface/flexibility)	Only method with genuine wave+current on a bed; Deltares covers winnowing/falling apron on sand.
Cable berm geometry & reshaping/deformation (any)	Deltares RoBeD (2023)	Van Rijn (D50 sizing)	Provides berm geometry + deformation class rather than only D50.
Quick first sizing / sanity check	Hudson / Van der Meer	—	Classical and fast, but without wave period or current.

5.9 Thusyanthan et al. (2013)

Thusyanthan, Jegandan and Robert (2013) is a paper on rock berm stability that is often used by companies and regulators in industry practice. Because it is well known, it is discussed here, even though it is not one of the six core methods compared in this thesis (Section 5.8).

The paper gives two methods for rock berm stability, one based on Soulsby (1997) and one based on the CIRIA Rock Manual (2007), plus design charts for a flat seabed and an amplification factor for berms with 1:3 or 1:4 side slopes. This method is left out of the comparison for three reasons.

First, it does not use a scalable damage level. Both methods are based on the threshold of motion, not on a parameter that lets the designer choose how much movement is acceptable, like Sd (Van der Meer), r (Van Rijn), or Nod (Van Gent and Van der Werf, Herrera and Medina). This means the method is calibrated for little to no movement, which tends to give larger, more conservative stone sizes.

Second, breaking waves are not properly addressed. The word "breaking" only appears as a label on the design charts, with no discussion of how the formulae behave near or after breaking, or up to what point they stay valid.

Third, the treatment of wave-current direction is inconsistent. The Soulsby-based method includes an angle between wave and current direction in its formula (Eq. 11), but the design charts do not vary or report this angle, so they likely assume the most conservative case, waves and current in the same direction. The CIRIA-based method has no directional term at all.

Together, these three points mean the method is more conservative than the methods used in this thesis, fitting its purpose as a quick, safe check for early-stage engineering rather than a detailed design tool. Since avoiding unnecessary over-design is a key goal of this thesis (Chapter 1), this method is discussed for completeness but not used further. Instead, Van Rijn (2019) and the other methods already compared in Section 5.8 are considered more suitable for the situations in this thesis, since they allow the required stone size to be tuned to an explicitly chosen damage level rather than defaulting to the threshold of motion.

- Schiereck & Verhagen (2019)

Ch. 8.2 (Stability general)
Ch. 8.3 (Stability of loose grains, slope stability under wave loading)

- Van Rijn (2019) — full paper
- Van Gent & Van der Werf (2014/ICCE2014) — full paper
- Herrera & Medina (2015) — full paper
- Deltares Handbook (2023)

Ch. 4 (Design of loose rock scour protections)
Ch. 5 (Loose rock berms at cable crossings)

- Thusyanthan, N.I., Jegandan, S., Robert, D.J. (2013), "Stability of Rock Berm under Wave and Current Loading", Proceedings of the 23rd International Offshore and Polar Engineering Conference (ISOPE), Anchorage, Alaska

6. Scour

6.1 The Scour Process

Scour is the local or large-scale erosion of the seabed caused by the flow disturbance introduced by a structure, or by natural morphodynamic processes. Two spatial scales are distinguished.

Local scour develops around individual structural elements, and around discrete, blunt obstructions such as piles it is characteristically driven by a horseshoe vortex that forms where the approaching flow separates at the base of the structure, together with flow acceleration and wake turbulence, it typically develops relatively quickly. A low, sloped structure such as a rock berm does not present the same blunt, vertical face, so this specific mechanism is less relevant there. Edge scour around a berm (Section 6.3) is instead driven by flow acceleration and locally amplified turbulence at its boundary.

Global scour, or global bed level lowering, is a broader depression around an entire structure or footprint, or results from large-scale morphodynamics unrelated to any single structure. It generally takes considerably longer to develop than local scour, and the two are not mutually exclusive. A structure can show both a local scour hole and a wider global lowering around it at the same time (Deltares HaSPro Handbook, 2023, Ch. 2).

Scour development also depends on whether sediment is supplied from upstream. In clear-water scour, there is no sediment transport into the scour hole from upstream, either because the upstream velocity is below the threshold of motion or because the upstream bed is not erodible. The scour hole develops relatively slowly and its final depth is sensitive to the exact flow conditions.

In live-bed scour, sediment is supplied from upstream but the local transport capacity around the structure is larger, so material is removed faster than it is replenished, live-bed scour reaches its equilibrium depth considerably faster than clear-water scour, though the depth of the hole then fluctuates as bedforms migrate through it (Schiereck & Verhagen, 2019, Ch. 4). The maximum scour depth generally occurs near the transition between the two regimes, and this transition is what the empirical scour-depth formulas in the literature are calibrated to predict.

6.2 Scour Around Subsea Cables

Most scour research and design guidance, including the depth predictions referred to above, has been developed for compact structures such as monopiles, jacket legs, or breakwater heads. A subsea cable is physically a very different object: elongated and narrow relative to its length, lying directly on or just below the seabed rather than penetrating it as a pile does. Flow disturbance around a cable is therefore closer to that around a long cylinder lying on the bed than to that around a vertical pile. A horseshoe vortex can still form where flow separates in front of the cable, similar to a pipeline, but it develops along the length of the cable rather than as a single localised feature. The resulting scour pattern is a trench-like depression that can run continuously along sections of the route rather than as a localised pit at a single point.

The direct consequence for structural integrity is that scour beneath a cable can cause a section of cable to become unsupported, or free-spanning, over the resulting depression. A free span introduces bending and, where currents or waves generate oscillating flow past the span, the possibility of vortex-induced vibration and fatigue loading that a fully supported cable does not experience.

DNV-RP-F109 (2021, Section 8.1) adds an important nuance here: a free span does not only introduce risk. The hydrodynamic lift force is actually reduced in a free span, because water can flow between the seabed and the cable, which can make the span more laterally stable rather than less, at least where friction governs the soil resistance, as on sandy or rocky seabeds. On clay, the loss of passive soil resistance in the span can have the opposite effect. Once vortex-induced cross-flow vibration sets in, however, the drag force increases significantly, and DNV-RP-F105 is the relevant reference for assessing this in detail. Because a rock berm protecting a cable is itself an elongated structure with the same alignment as the cable, edge scour along the berm (Section 6.3) and the free-spanning behaviour of the cable underneath it are closely related, rather than two separate problems.

6.3 Scour and Berm stability

Once a section of seabed is protected by a rock berm, scour does not stop, it shifts to the edges of the protected area. These edge scour holes form because the presence of the berm itself locally amplifies the flow and increases turbulence at its boundary, in a manner similar to local and global scour around an unprotected structure. Edge scour holes are generally smaller than local scour holes around an unprotected structure and take longer to develop, but in a tidal environment they tend to develop on alternating sides of the berm as the current reverses, since each reversal partially backfills the hole formed during the previous phase (Deltares HaSPro Handbook, 2023, Ch. 2).

This has a direct consequence for berm design: if the edges of a rock berm are undermined by scour, the berm can lose its foundation support at the margins and settle or slump into the resulting hole, regardless of whether the armour stones on top of the berm are individually stable under the design wave and current loading. Scour and berm stability are therefore coupled and cannot be designed independently: a berm sized correctly for hydraulic stability (Chapter 5) can still fail if its edges are not adequately protected against, or given sufficient allowance for, ongoing edge scour. Extending the berm in the direction of the dominant flow, so that a larger area of erosion-resistant material covers the zone where edge scour would otherwise develop, is one way this coupling is addressed in practice. DNV-RP-F109 (2021, Section 8.2) adds a direct warning relevant to this kind of intermittent protection: where support along a route is not continuous, scour can develop into a free span. This reinforces why edge scour at the boundary of a berm cannot be treated as a secondary or cosmetic issue.

Deltares HaSPro Handbook (2023, Section 5.2.3.1) explicitly states that quantitative guidelines for edge scour around rock berms are still lacking, despite considerable research into the underlying mechanisms. This is treated in this thesis as a genuine, currently unresolved gap in the state of the art, rather than a simplification specific to this study, and is revisited as an open point in Chapter 8.

Scour around a cable is not always purely detrimental. DNV-RP-F109 (2021, Section 8.5) notes that local scour can also cause a cable to self-lower and partially self-bury over time, which is a benign effect rather than a threat, since it increases the effective cover without any construction effort. This should only be relied upon with an appropriate safety margin, and the handbook gives specific conditions under which it should not be assumed: the mobile bed must extend along the whole relevant section of cable, self-lowering is reduced or prevented where more than 10% of the bed material by mass is finer than 75 μm, and hard points such as large rocks beneath the seabed can prevent further lowering regardless of how mobile the surrounding bed is.

6.4 Scour Mitigation Approaches

Loose rock protection, in the form of a graded rock berm, is the primary scour and cable protection method considered in this thesis, and is treated in detail in Chapter 5. The Deltares HaSPro Handbook (2023) also describes several alternative scour protection systems that are used in practice as either an alternative to, or in combination with, loose rock:

Artificial vegetation systems consist of a dense canopy of individual synthetic fronds attached to an anchored frame of ballasted mattresses. They work by damping near-bed turbulence and reducing flow velocities at the bed rather than by armouring it directly, and their design considerations include both overall stability and the prevention of edge scour around the mattress frame.

Block, gabion, and ballast-filled mattresses are weighted mattress systems placed directly on the seabed. Block mattresses use interlocking concrete units, gabion mattresses use rock- or gravel-filled wire mesh compartments, and ballast-filled mattresses use a flexible fabric envelope filled with a ballast material; all three are used to protect pipelines and cables as well as structures, and, like loose rock, remain susceptible to edge scour at their perimeter if that is not separately addressed.

DNV-RP-F109 (2021, Section 8.2) lists a further set of measures used in practice for on-bottom stability more generally, including weight coating, trenching, burial or intermittent burial, structural anchors, grout bags, and big bags, alongside mattresses and intermittent rock berms. These are noted here for completeness rather than evaluated in detail, since they address the cable's own on-bottom stability rather than scour protection specifically.

Each of these alternative systems has different installation requirements, flexibility to accommodate seabed movement, and edge-scour behaviour, and the Deltares Handbook indicates suitability per situation rather than a single preferred system. For the scope of this thesis, loose rock berms remain the primary method considered, since they are the most established and most directly comparable across the five core sources; the alternative systems above are noted here as context for situations where a loose rock berm may not be the preferred or feasible solution.

6.5 Anchor and Trawling Protection

Alongside hydrodynamic loading, a rock berm intended to protect a subsea cable may also need to withstand impact loads from bottom-trawling fishing gear and dragging ship anchors. These loads are not computed as part of the design tool in this thesis, since including them by default would risk significant overdesign. They are treated here only as boundary conditions to be aware of. It should also be stressed that the required level of protection against these man-made hazards is not a fixed, universal value: it is determined per project through a site-specific risk assessment (a Cable Burial Risk Assessment, following DNV and CarbonTrust guidance), based on the actual shipping density, fishing intensity, and anchorage activity along that particular section of route. The figures given below are therefore illustrative of common industry practice, not fixed design rules that apply everywhere.

For trawling gear, industry practice (Van Kessel & Stam, 2010; CIRIA, 2007) has shown that a minimum rock cover of 0.50 m over the cable is sufficient to protect against dragging trawl boards and beams in effectively all cases, with typical trawl board weights of 500–2000 kg at speeds of 3–5 knots corresponding to impact energies of 0.5–6 kNm; the slope of the berm deflects the trawl board so that only part of this energy needs to be absorbed, and penetration into the berm remains below about 0.30 m. This 0.5 m figure is a widely used project baseline in areas with active bottom trawling, but the actual required cover is still assessed per project against local fishing gear and activity.

Protection against a dragging ship anchor is a different matter, and even more strongly project- and location-dependent. Van Kessel and Stam (2010) show that rock berm dimensions designed to force a large anchor to break out and re-surface, rather than cut into or under the berm, scale directly with anchor size: for a representative 3-tonne stockless anchor, their rules of thumb already indicate a berm with a bottom width of around 15 m. Because anchor-related berm requirements can become disproportionately large compared to the wave-driven design considered in this thesis, computing anchor protection by default would directly conflict with the "avoid overdesign" objective of this thesis. Consistent with industry guidance, the recommended approach where large dragging anchors pose a credible risk is to route the cable away from high-risk areas (shipping lanes, anchorages, fishing grounds) rather than to design the berm for the largest conceivable anchor impact; anchor protection is therefore noted here as a boundary condition, to be sized on a per-project basis according to the identified risk, and as a possible follow-up study, not as part of the computational design tool developed in Chapter 7.

- Van Kessel, R., Stam, C.J. (2010), "The Lights Go Out: The Ultimate Protection Technology for Protecting Submarine Cables", SubOptic 2010.
- CIRIA (2007), The Rock Manual: The Use of Rock in Hydraulic Engineering, 2nd edition, CIRIA C683
- Schiereck & Verhagen (2019) Ch. 4 (Flow-Erosion: scour as sediment transport, scour process, scour with bed protection, equilibrium scour, stability of protection)
- Deltares Handbook (2023), Ch. 2 (Scour prediction), Ch. 3 (Scour mitigation strategies A/B/C), Ch. 5 (Loose rock berms at cable crossings), Ch. 9–11 (Alternative systems)

- Schiereck & Verhagen (2019), Ch. 4 (Flow-Erosion: scour as sediment transport, scour process, scour with bed protection, equilibrium scour, stability of protection.
- Deltares Handbook (2023), Ch. 2 (Scour prediction, general description of the scour process, global and local scour, edge scour), Ch. 9 (Alternative scour protection systems), Ch. 10 (Artificial vegetation), Ch. 11 (Block, gabion and ballast-filled mattresses)
- DNV-RP-F109 (2021), Section 8.1 (Free spans), Section 8.2 (Mitigating measures), Section 8.5 (Pipeline on mobile seabed)

7. Design Framework & Computational Approach

7.1 Input Parameter Space

All parameters required for a stability assessment, grouped by category: hydrodynamic (Hs, Tp, Tm-1,0, uc, wave direction), geometric (h, ht, bed slope), material (Dn50, ρs, D50) and berm geometry (Bt, tt). For each parameter, physical meaning, unit, and typical nearshore range.

7.2 Core Design Formulae

All formulae that will be implemented in the Python script, written out in full with variable definitions.

This chapter is the technical foundation of the script, every formula here maps directly onto the code.

Organization per method so its well-structured for the code. Like Chapter 6?.

7.3 Failure Criteria & Damage Levels

Stability for each method.

Shields with a threshold ($\theta \leq \theta cr$).
Van Rijn with damage classes (r = 0.4 to 1.0).
Van Gent NOD with an acceptable upper bound.

I have to make these differences explicit so they can be implemented consistently in the script.

7.4 Validity Ranges & Applicability Warnings

State the validity limits per method (e.g. Van Gent: ht/h = 0.10–0.30, h/Hs = 1.2–4.5). In the script these become the warnings issued to the user when input falls outside the valid range. This is essential for a tool intended for global application.

7.5 Sensitivity Analysis Framework

Sensitivity analysis as a methodology: Identifying which input parameters most strongly influence the outcome.

Which methods do I want to apply

local sensitivity analysis and global methods such as Sobol indices and the Morris screening method.

Thesis methodology chapter.

- Schiereck & Verhagen (2019)

Ch. 3.2 (Stone dimensions, waterdepth, practical application)
Ch. 10 (Dimensions, probabilistics, failure mechanisms)

- Van Rijn (2019)

Section 2–4 (all core formulae and validity ranges)

- Van Gent & Van der Werf (ICCE2014)

Equations 1–3 and validity discussion

- Herrera & Medina (2015)

Equations 1–4 and validity ranges

- Deltares Handbook (2023)

Ch. 4.4 (Design approach, external stability, interface stability, flexibility)
Ch. 4.5 (Design workflow, design parameters)

Output must be for every little part of the cable route what options are there for constructing the cross-section. So if we use this d50 we need to use this slope, if we use this d50 we need to use this slope.

Beyond sizing the armour stone (Chapter 5) and computing the berm geometry (Section 5.6), the Deltares HaSPro Handbook (2023, Section 5.2.4) recommends three steps for dimensioning a rock berm as a cable cover. This thesis uses the same three steps to structure the design tool.

Step 1, external stability: calculate the minimum berm size needed for hydraulic stability, using the RoBeD-based TOPC calculation from Section 5.6 (Deltares HaSPro Handbook, 2023, Eq. 20-26).

Step 2, interface stability (winnowing check): check that the installed berm height is at least four rock layers. If it is not, extra measures are needed, such as a higher berm or a monitoring plan. This check is simple and does not depend on any unverified assumptions, so it is included directly in the design tool.

Step 3, flexibility (edge-scour check): check whether the berm can follow expected changes in the seabed level at its edges. The Deltares HaSPro Handbook (2023, Eq. 27, after Van Velzen, 2012) gives a formula for the extra rock volume needed to act as a falling apron:

$$V_{apron} = \gamma_s \cdot D_{50} \cdot (n_{D,top} + n_D) / (2 \cdot \sin\alpha) \cdot h_{tot}$$

Here γ_s is a safety factor for imperfect launching and 3D effects, $n_{D,top}$ and n_D are the layer thickness, in number of rock layers, at the top and toe of the falling apron slope, α is the slope angle of the falling apron, and h_{tot} is the total expected bed lowering.

This formula needs h_{tot}, the expected bed lowering, as an input. The Deltares HaSPro Handbook (2023, Section 5.2.3.1) states that no validated method currently exists to predict edge scour depth around rock berms specifically. Because of this, h_{tot} cannot be obtained from a reliable formula for this application, and any value used would be little more than a guess. For the same reason the anchor and trawling loads in Section 6.5 are not computed by default, this thesis does not include the falling apron formula in the design tool. Adding it would create a false sense of precision instead of a real one. The formula and its role are described here so the flexibility requirement is not ignored, and computing it properly is left as an open point for future work (Chapter 8).

8. Knowledge Gaps & Research Questions

Conclusion of theliterature review.

What do I know:

1. stability criteria for uniform flow are well developed,
2. the Deltares HaSPro approach is state-of-the-art for offshore applications,
3. and several methods exist for toe and berm stability.

What we do not know:

1. there is no validated, integrated method specifically for nearshore cable berms under combined wave-current conditions,
2. and existing methods are frequently applied outside their validity ranges without adequate justification.

I have to formulate the research questions that follow directly from the literature, which justify my methodology chapter and the development of my Python tool.

Rather than treating the six stability methods as six competing options to choose between, I've organized the tool around three stages that reflect what each type of method is actually used for.

Stage one is obtaining a candidate rock size. Hudson, Van der Meer, Van Rijn, and Van Gent & Van der Werf all belong here, each takes the site conditions and gives back a required stone size, at some assumed level of accepted damage.

Stage two is comparing that rock size against the level of damage it implies, and making an actual decision. Rather than just accepting whatever single answer the tool produces, the idea is to show the trade-off, how the required stone size changes as the accepted damage level changes, across a range, so the choice of "how conservative do we want to be here" is visible and deliberate, not hidden inside a default value. Herrera & Medina fits here specifically, since its formula doesn't give a stone size directly at all, it gives you damage for a candidate size, so you only get an answer by exploring exactly this kind of trade-off.

Stage three is finalizing the design with the Deltares HaSPro/RoBeD model. Once a rock size and an accepted damage level have been chosen, Deltares takes that and produces the actual buildable berm, height, required cover, and whether interface stability and flexibility hold up over time.

I think this three-stage structure is a useful way to present the design framework in the thesis, it makes clear that the tool isn't just running six formulas and picking one, it reflects a genuine engineering decision process, size it, evaluate the trade-off, finalize it.

Where this structure runs into a real problem:
The issue sits exactly between stage two and stage three. Each method expresses "acceptable damage" differently, an implicit fixed level for Hudson, a damage number Sd for Van der Meer, a reduction coefficient r for Van Rijn, a damage number Nod for Van Gent and Herrera & Medina, and deformation classes 0 through 3 for Deltares. These aren't the same physical quantity, and there's no established formula or table converting one into another, they come from different test setups and different structures.

So whatever damage level someone accepts in stage two is expressed in that method's own language, but stage three, Deltares, needs a deformation class, not an r or a Nod. There's currently no justified way to carry the decision from stage two into stage three, without either inventing an unbacked conversion, or asking for a second, disconnected decision about acceptable deformation class that has no explicit link to what was already decided.

This is already flagged as an open question in my project scope, mapping r, Nod, and deformation class onto a common axis is something still to be agreed, not something I've solved. My options as I see them:

- Option 1: I could just keep the two decisions separate and not try to link them. I pick my rock size using whichever method applies, and I'm happy with the damage level that method already uses. Then, separately, when I get to Deltares, I just choose a deformation class myself, without calculating it from the first choice. Two independent decisions, not one flowing into the other.

- Option 2: I could come up with my own rough conversion between the two, but say clearly that it's my own proposal, not something from the literature. So I'd say something like, "I'm treating this damage level as roughly equivalent to this deformation class," based on my own comparison, and in the thesis I'd be upfront that this needs validation later, it's a contribution I'm making, not a fact I found in a paper.

- Option 3: I could only connect the methods where a link genuinely makes sense, and leave the rest disconnected. So I'd connect Van Rijn to Deltares, since we already know those two are meant to work together for the sand case, but for a method like Herrera & Medina, where the link would be forced, I'd just present its result on its own, without trying to feed it into Deltares at all.

https://www.researchgate.net/publication/235634121_Experimental_Investigation_of_Water_Wave_Characteristics_in_a_Wave_Channel/figures?lo=1

Table 5.8:

Hudson (1953/59) — *Source*: CIRIA C683 Rock Manual (2007), §5.2.2.2 (Eq. 5.133–5.135) and Table 5.29 (p.579); Schiereck & Verhagen (2019), §8.3.2, Eq. 8.10.

> VERIFIED against CIRIA. CIRIA gives NO strict cotα validity range for Hudson (unlike Van der Meer). Derived from regular-wave tests on permeable-core structures. KD = 4 (non-breaking) / 2 (breaking, per SPM 1984); impermeable core only with KD = 1. Per Table 5.29 Hudson is applicable for deep to shallow water but NOT very shallow (Hs-toe < 70% Hso). Your earlier 1.5–4 is not in CIRIA; use Schiereck's 1:1.5–1:6 or drop the upper bound.

Van der Meer (1988/98) — *Source*: CIRIA C683 Rock Manual (2007), Table 5.24 (p.570, deep-water validity) and Table 5.29 (fields of application); Schiereck & Verhagen (2019), §8.3.2–8.3.3.

> VERIFIED against CIRIA Table 5.24 (authoritative). Deep water (h > 3·Hs-toe): cotα = 1.5–6, N < 7500, som = 0.01–0.06, ξm = 0.7–7, Δ = 1–2.1 (for Δ up to 3.5 → cotα ≥ 2), P = 0.1–0.6 (P = 0.1 impermeable, 0.4 armour+filter on permeable core, 0.5 armour on permeable core, 0.6 homogeneous). Plunging/surging split. The shallow-water form (Van Gent et al. 2004, needs H2%) covers very shallow water only and is OUT OF SCOPE for this thesis.

Van Rijn (2019) — *Source*: van Rijn (2019), Critical movement of large rocks in currents and waves, Eq. 3 (θcr), Eq. 5 (wave+current), Table 1 (fc/fw ranges); Cheng (2002) for the dynamic case.

> VERIFIED against the paper. Added vs your table: roughness ratio h/(αD50) & Aw/(αD50) = 10–300, and D50 is the sieve diameter with D50 ≈ 1.2·Dn50 (convert if inputs are nominal Dn50).

Van Gent & Van der Werf (2014) — *Source*: van Gent & van der Werf (2014), Rock toe stability of rubble mound breakwaters, validity statements in the paper (ht/h and h/Hs ranges, tested steepnesses).

> VERIFIED against the paper. Added vs your table: wave steepness sp = 0.015–0.04 (tested), non-overtopped structure, and the tests used a fixed (non-mobile) bed so toe scour was excluded. Uses Tm-1,0 (spectral period) and Hs = Hm0.

Herrera & Medina (2015) — *Source*: Herrera & Medina (2015), Toe berm design for very shallow waters on steep sea bottoms, Eq. 12; validity ranges §2; damage levels after CIRIA Rock Manual (2007).

> VERIFIED against the paper (pages you sent). -0.15 < hs/Hs0 < 1.5 is correct (negative lower bound = emerged berm). Damage levels per CIRIA (2007): Nod = 0.5 (start), 2.0 (moderate), 4.0 (failure). REMOVED "Nod = 1 = design" – it is not in Herrera's cited criteria. CORRECTED steepness: Herrera's own validity table (p.10, column "This study") gives 0.008–0.08, not 0.02–0.07, and Ns = 0.81–3.36 (verify the exact steepness symbol s0p vs stp when you cite it). SCOPE NOTE: Herrera's domain (very shallow, m=1/10, heavy breaking) sits at the edge of or just outside this thesis's nearshore domain, so it will rarely be the governing method here.

Deltares HaSPro / RoBeD (2023) — *Source*: Deltares (2023), Handbook of Scour and Cable Protection Methods, §5.2–5.3 (RoBeD, p.73–75), Appendix A (mobility number, Eq. A.1 & A.6). Polynomial coefficients themselves are in the loose-rock analysis report Deltares (2023n), which is not in our set.

> The handbook's RoBeD section explicitly recommends MOBthresh = 0.95 for the rock-berm model, so "no fixed default" understates the source. MOB = θ/θcr; θcr from Soulsby (Eq. A.6, which keeps the 1.2 factor) and ks = 2.5·D50 for the mobility calc – note these differ from the Van Rijn (2019) θcr form used elsewhere.